Advantest Europe GmbH

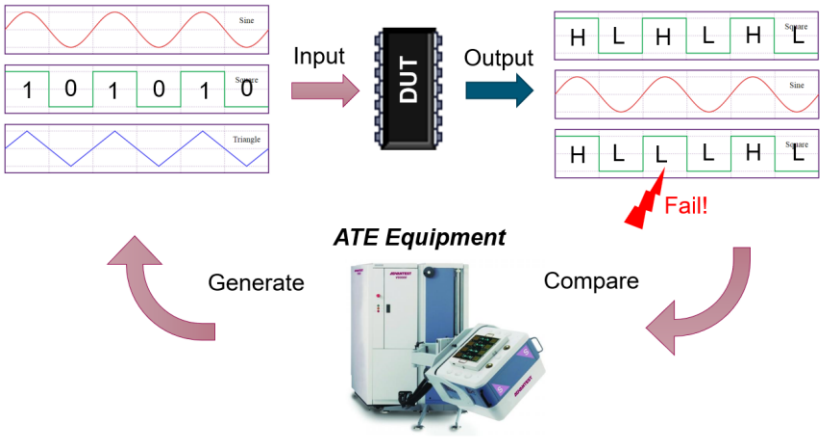
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Challenge X: Compressing/Reducing pattern vector data content

1 WHAT THE CHALLENGE IS ABOUT:

In ATE test a semiconductor tester applies a stimulus to the DUT (Device under Test), so for digital pins this is usually a stream of 0 or 1. The ATE test system then checks the DUT output signals against the expected output values of L/H. Only if all output values come out as expected, the DUT is considered as pass, i.e. working. The input values to the device and the expected output values are usually stored in the form of a pattern, as shown below:



Example Pattern for a counter

Reset	Clock	Load	Enable P	Enable T	Data Input P0	Data Input P1	Data Input P2	Data Input P3	Data Output Q0	Data Output Q1	Data Output Q2	Data Output Q3	Carry-Out
0	1	0	0	0	0	0	0	0	L	L	L	L	L
1	1	0	0	0	0	1	0	1	L	H	L	H	L
1	1	0	0	0	1	0	1	0	H	L	H	L	L
1	1	0	0	0	0	0	0	0	L	L	L	L	L
1	1	1	1	1	0	0	0	0	H	L	L	L	L
1	1	1	1	1	0	0	0	0	L	H	L	L	L
1	1	1	1	1	0	0	0	0	H	H	L	L	L
1	1	1	1	1	0	0	0	0	L	L	H	L	L
1	1	1	1	1	0	0	0	0	H	L	H	L	L
1	1	1	1	1	0	0	0	0	L	H	H	L	L
1	1	1	1	1	0	0	0	0	H	H	H	L	L
1	1	1	1	1	0	0	0	0	L	L	L	H	L
1	1	1	1	1	0	0	0	0	H	L	L	H	L
1	1	1	1	1	0	0	0	0	L	H	L	H	L
1	1	1	1	1	0	0	0	0	H	H	L	H	L
1	1	1	1	1	0	0	0	0	L	L	H	H	L
1	1	1	1	1	0	0	0	0	H	L	H	H	L
1	1	1	1	1	0	0	0	0	L	H	H	H	L
1	1	1	1	1	0	0	0	0	H	H	H	H	H
1	1	1	1	1	0	0	0	0	L	L	L	L	L
1	1	1	0	1	0	0	0	0	L	L	L	L	L
1	1	1	1	0	0	0	0	0	L	L	L	L	L

The patterns can be simple functional patterns as shown in the example here for a counter. Observe how the Data output values increase on each clock cycle. Most pattern test data are SCAN test data created by Design-For-Test tooling. Due to Moore's law the number of transistors on a chip are doubling every two years. Since the amount of SCAN data is scaling linearly with the number of transistors, the overall scaling is again exponentially. Thus, the amount of vector memory in ATE Digital instruments also needs to grow exponentially to support the testing of future chips. One digital instrument now has 0.5TB of DRAM memory to store the pattern vectors, one tester can contain 20 digital instruments, leading to an overall total amount of memory in an ATE tester being in the range of 10 TB!

Handling up to 10 TB of pattern data (compressed typically ~0.5 TB) leads to many challenges:

- Storing that amount of data on central file-shares as part of the overall Test-Program
- Transferring the data over the network to the tester workstation SSD (sometimes over slow 100 Mbit factory networks)

- Storing multiple test-program versions on tester workstation SSD outgrows the available space.
- Downloading the pattern data from the tester workstation into the actual HW instrument DRAM memory takes minutes.

Being able to reduce/compress the amount of data will help in all these cases. The data today is already getting compressed using the zstd algorithm, so you will need to find more innovative ways to compress redundant data.

2 What the teams have to do:

Understand the pattern vector format of an ATE test-system and the relevant boundary conditions. Come up with possible data compression ideas and prototypes using the sample pattern vector data provided. Benchmark the results of the different compression ideas, regarding compression factor and compression/decompression speed.

3 What the teams produce:

Prototype implementations in the programming language of their choice. Benchmark results presentation. A list of approaches tried which did not work out.

4 Tools required:

Nothing required, except a personal laptop with the programming tools of choice installed. We'll provide an initial sample corpus of pattern vector data. In addition, we'll provide access to our online ATE test-system documentation center to help understanding the pattern vector data format and the relevant boundary conditions.

5 How it is judged, and the outcome:

At the 30h mark we will provide a bigger corpus of pattern vector test data. Teams can then demonstrate their compression factor improvements on a bigger corpus of data and can test how their compression/decompression speeds scale.

6 Scoring criteria

Every team is scored on the same five criteria. Each criterion is scored from 0 to 5. The challenge-specific criterion counts double (2x), so the maximum is 30 points.

Criterion	Weight	What judges look at
Technical Feasibility	1x	Does the solution work, and is the approach technically sound? Judges look for a working prototype or a credible technical path, and for evidence that is measured, reproducible and honest about its limits.
Innovation	1x	How new is the idea or approach compared with the current state of the art, and how creative is the way it tackles the problem?
Market Potential	1x	Is there a real need, a clear customer or user, and room to grow? Judges look at how large and how reachable the opportunity is.
Business Viability	1x	Could this become a sustainable product or business? Judges look at the business model, the cost and effort of adoption, the route to market and what it would take to get there.
Challenge-specific: Compression results	**2x**	Compression factor and compression/decompression speed on the larger pattern vector corpus provided at the 30h mark, including how the speeds scale.

Total: 6 weighted units (1 + 1 + 1 + 1 + 2), maximum 30 points.

CHALLENGE-SPECIFIC CRITERION: COMPRESSION RESULTS (2X)

• Compression factor achieved compared with the data as compressed today, shown on the bigger corpus provided at the 30h mark.

• Compression and decompression speed, and how both scale on the bigger corpus.

• Benchmark results presented for every idea tried.